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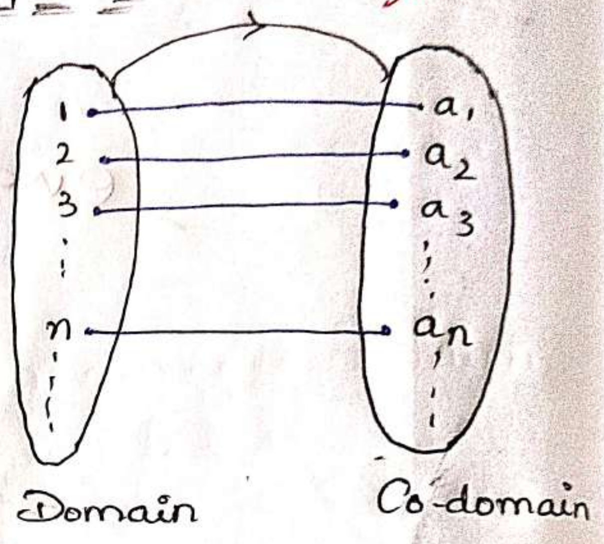
Unit-4 Recurrence Relations

1. Explain the concept of a number sequence and recurrence relations for numbers sequences. Using these explanations, derive the general nth term formulas (closed-forms) and recurrence relations corresponding to: (i) Arithematic sequence. (ii) Geometric sequence, and (iii) Fibonacci sequence

Sol: **Sequences or Number Sequences.**

Let $N = \{1, 2, 3 \ldots\}$ be the set of natural numbers and let $\mathbb{R}$ be the set of real numbers. Then any of formation from N to $\mathbb{R}$ is called "sequence". Generally, it is denoted by



$$\boxed{a_n : N \to \mathbb{R}}$$

and represented by $\{a_n\}_{n=1}^{\infty}$ or simplify $\{a_n\}$, where a_n is called nth term of sequence $\{a_n\}$.

Suppose $W = \{0, 1, 2, 3, 4 \ldots\}$ be the set of whole numbers then the function $a_n : W \to \mathbb{R}$ is represent by $\{a_n\}_{n=0}^{\infty}$

Example:

Sequence	n^{th} term
$\{a_n\} = \{1, 1, 1 \ldots\}$	$a_n = 1$
$\{a_n\} = \{1, \frac{1}{2}, \frac{1}{3} \ldots\}$	$a_n = \frac{1}{n}$
$\{a_n\} = \{-1, 1, -1, 1, -1\}$	$a_n = (-1)^n$
$\{a_n\} = \{0, 2, 0, 2 \ldots\}$	$a_n = 1 + (-1)^n$
$\{a_n\} = \{\frac{1}{2}, \frac{2}{3}, \frac{3}{4} \ldots\}$	$a_n = \frac{n}{n+1}$

Recurrence Relation

Let $\{a_n\}$ be any Sequence of numbers, then the mathematical expression

$$a_n = f(a_{n-1}, a_{n-2}, a_{n-3}, \ldots, a_{n-k}, n)$$

(or)

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \ldots + c_k a_{n-k} + f(n)$$

is called reccurrence relation for $\{a_n\}$, where $c_1 \ldots c_k$ are Constants and f is any function on n.

1. Arithematic Sequence

$a_1 = a,$

$a_2 = a + d,$

$a_3 = a + 2d,$

$\vdots$

$a_{n-1} = a + (n-2)d$

$a_n = a + (n-1)d \Rightarrow a + [(n-2) - 1]d \Rightarrow a + (n-2)d - d$

$\Rightarrow a_{n-1} - d$

$\vdots$

$\therefore$ The arithematic Sequence is $\{a_n\}$, and its Corresponding recurrence relation is $\boxed{a_n = a + (n-1)d}$

(or) $\boxed{a_n = a_{n-1} - d}$

2. Geometric Sequence

$a_1 = r,$

$a_2 = ar,$

$a_3 = ar^2,$

$a_4 = ar^3$

$\vdots$

$a_{n-1} = ar^{n-2}$

$a_n = ar^{n-1} \Rightarrow (ar^{n-2}).r^{+1} = a_{n-1}.r$

$\therefore$ The recurrence relation for the geometric Sequence $\{a_n\}$ is

$$\boxed{a_n = r \cdot a_{n-1}}$$

3. Fibonacci Sequence

$f_0 = 0$

$f_1 = 1$

$f_2 = 1 \Rightarrow (0+1)$

$f_3 = 2 \Rightarrow (1+1)$

$f_4 = 3 \Rightarrow (1+2)$

$f_5 = 5 \Rightarrow (2+3)$

$\vdots$

$f_{n-1} = f_{n-2} + f_{n-3}$

$f_n = f_{n-1} + f_{n-2}$

$\vdots$

The fibonacci Sequence recurrence relation of the fibonacci sequence $\{f_n\}$ is

$$\boxed{f_n = f_{n-1} + f_{n-2}}$$

Name of the Sequence	Closed form	RR of the Sequence.
Arithematic Sequence	$a_n = a+(n-1)d$	$a_n = a_{n-1}+d$
Geometric Sequence	$a_n = r.a^{n-1}$	$a_n = r.a_{n-1}$
Fibonacci Sequence	$f_n = f_{n-1} + f_{n-2}$	$f_n = f_{n-1} + f_{n-2}$

Solving methods for RRs.

- Substraction method
- Iteration method
- Characteristic equation method
- Generating function method.

2. Let $\{a_n\}$ be defined by recurrence relation

(i) $a_n = a_{n-1} + 3$, with $a_0 = 2$ (4)

(ii) $a_n = 2a_{n+1}$, with $a_1 = 1$

(a) Find the Values of a_1, a_2, a_3.

(b) Using the method of ~~repeated~~ method for Solving Substitution obtain a closed form expression for a_n.

Substitution Method

Step 1: The given RR

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots c_k a_{n-k} + f(n) \rightarrow (1)$$

with initial Condition a_0.

Step 2:

Use a_0 to find a_1

Step 3:

Use a_1 to find a_2

⋮

Step n:

Use a_{n-1} to find a_n.

Here a_n is the Solution of (1)

(i) The given RR is

$$a_n = a_{n-1} + 3 \rightarrow (2) \text{ with } a_0 = 2$$

For $n=1$, (2) implies

$$a_1 = a_0 + 3$$

For $n=2$, (2) implies

$$a_2 = a_1 + 3 = (a+3) + 3$$

For $n=3$, (2) implies

$$a_3 = a_2 + 3$$

$$a[(a+3)+3] + 3$$

By the Substitution method,

$$a_n = a_0 + (3 + 3 + \cdots + 3) \text{ (n terms)}$$

$\Rightarrow a_n = a_0 + 3(1+1+ \cdots +1)$ (n terms)

$\Rightarrow a_n = a_0 + 3n$ ⑤

$\Rightarrow \boxed{a_n = 2+3n}$. since $a = 2$

It is the Solution (closed form) of ②

(ii) The given RR is

$$a_n = 2a_{n-1} + 1 \longrightarrow ③ \text{ with } a_1 = 1$$

For $n = 2$, ③ implies

$$a_2 = 2a_1 + 1 \quad (a+1)+1$$

For $n = 3$, ③ implies

$$a_3 = 2a_2 + 1 = 2(2a_1 + 1) + 1$$

For $n=4$, ③ implies

$$a_4 = 2a_3 + 1$$
$$= 2[2(2a_1+1)+1]+1$$

⋮

By the Substitution method.

$$a_n = 2^{n-1}a_1 + 2^{n-2} + 2^{n-3} - \cdots + 1$$

put $a_1 = 1$, we get

$$a_n = 2^{n-1} + 2^{n-2} + 2^{n-3} - \cdots + 2 + 1$$

∴ It is the Solution of ③

③ Describe the iterative method for Solving reccurence relations and ~~the~~ apply this method to solve:

(i) $a_n = -a_{n-1}$, $a_0 = 5$ (ii) $a_n = a_{n-1} + 1$, $a_0 = 1$

(iii) $a_n = a_{n-1} - n$, $a_0 = 4$ (iv) $a_n = 2a_{n-1} - 3$, $a_0 = -1$

Sol: Iteration Method

Step 1:- The given RR is

$$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k} + f(n) \rightarrow ①$$

, with ~~Iterative method~~ a_0

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Step 2: The given Solution (closed form) of ① is a_n

Step 3:- ⑥

~~The~~ To find $a_{n-1}, a_{n-2}, \ldots, a_{n-k}$ by using a_n,

Step 4:-

Substituting $a_n, a_{n-1}, a_{n-2}, \ldots, a_{n-k}$ in ①

We get

$\boxed{LHS = RHS \text{ of } ①}$

Hence, the given closest form a_n is the Solution of ①

(i) The given RR is

$$a_n = -a_{n-1} \rightarrow ② \quad \text{with } a_0 = 5$$

Replace n by $n-1$ in ②

$$a_{n-1} = -a_{n-2} \rightarrow ③$$

Replace n by $n-2$ in ②

$$a_{n-2} = -a_{n-3} \rightarrow ④$$

Replace n by $n-3$ in ②

$$a_{n-3} = -a_{n-4} \rightarrow ⑤$$

$\vdots \qquad \vdots$

by the iteration method

$$a_n = -a_{n-1}, \text{ by } ②$$

$$= -(-a_{n-2}), \text{ by } ③$$

$$= -(-(-a_{n-3})), \text{ by } ④$$

$$= -(-(-(-a_{n-4}))), \text{ by } ⑤$$

$\vdots \qquad \vdots$

$$a_n = (-1)^n a_{n-n}$$

$$a_n = (-1)^n \cdot a_0$$

$\boxed{a_n = 5(-1)^n}$ It is the Solution of ②

(ii) The given RR is

$$a_n = a_{n-1}+1 \rightarrow ② \text{ with } a_0 = 1.$$

Replace n by $n-1$ in ②

$$a_{n-1} = a_{n-2}+1 \rightarrow ③$$

Replace n by $n-2$ in ②

$$a_{n-2} = a_{n-3}+1 \rightarrow ④$$

Replace n by $n-3$ in ②

$$a_{n-3} = a_{n-4}+1 \rightarrow ⑤$$

$\vdots \qquad \vdots$

By the iteration method,

$$a_n = a_{n-1}+1, \text{ by } ②$$

$$= (a_{n-2}+1)+1$$

$$= ((a_{n-3}+1)+1)+1$$

$$= (((a_{n-4}+1)+1)+1)+1$$

$\vdots \qquad \vdots$

$$a_n = a_{n-n} + (1+1+1+\cdots+1\,(n \text{ terms}))$$

$$\Rightarrow a_n = a_0 + n$$

$\Rightarrow \boxed{a_n = 1+n}$ This is the solution of ②

(iii) The given RR is

$$a_n = a_{n-1} - n \rightarrow ② \text{ with } a_0 = 4$$

Replace n by $n-1$ in ②

$$a_{n-1} = a_{n-2} - (n-1) \rightarrow ③$$

Replace n by $n-2$ in ②

$$a_{n-1} = a_{n-3} - (n-2) \rightarrow ④$$

Replace n by $n-3$ in ②

$$a_{n-1} = a_{n-4} - (n-3) \rightarrow ⑤$$

By the iteration method,

$a_n = a_{n-1} - n$, by ②

$= (a_{n-2} - (n-1)) - n$ by ③

$= ((a_{n-3} - (n-2)) - (n-1)) - n$, by ④

$= (((a_{n-4} - (n-3)) - (n-2)) - (n-1)) - n$,

⋮

$a_n = a_{n-n} - (n-(n-1) + \dots + (n-2) + (n-1)) - n$

$\Rightarrow a_n = a_0 - (1 + 2 + \dots + (n-2) + (n-1)) - n$

$\Rightarrow a_n = a_0 + \frac{(n-1)((n-1)+1)}{2}$ $\left[\because 1 + 2 + \dots + n = \frac{n(n+1)}{2}\right]$

$\Rightarrow \boxed{a_n = 4 + \frac{(n-1)(n)}{2} + n}$ This is the solution of ②

(iv) The given RR is

$a_n = 2a_{n-1} - 3 \rightarrow$ ② with $a_0 = -1$

Replace n by $n-1$ in ②

$a_{n-1} = 2a_{n-2} - 3 \rightarrow$ ③

Replace n by $n-2$ in ②

$a_{n-1} = 2a_{n-3} - 3 \rightarrow$ ④

Replace n by $n-3$ in ②

$a_{n-1} = 2a_{n-4} - 3 \rightarrow$ ⑤

~~Replace n by~~

By the iteration method,

$a_n = 2a_{n-1} - 3$, by ②

$= 2(2a_{n-2} - 3) - 3$, by ③

$= 2(2(2a_{n-2} - 3) - 3) - 3$, by ④

$= 2(2(2(2a_{n-2} - 3) - 3) - 3) - 3$, by ⑤

⋮

$a_n = 2^n(a_{n-n} - (3+3+\cdots+3 \text{ (n terms)}))$

$\Rightarrow a_n = 2^n a_0 - 3n$

$\Rightarrow \boxed{a_n = 2^n(-1) - 3n}$ This is Solution of (2)

(4) Classify recurrence relations in terms of order, degree, linearity and homogeneity. and use this classification to find the order and degree of the following recurrence relations:

(i) $a_n = 3a_{n-1} + 4a_{n-2} + 5a_{n-3}$

(ii) $a_n = 2na_{n-1} + a_{n-2}$ and

(iii) $a_n = a_{n-1}^2 + a_{n-2}$.

Sol:- Order and Degree of the RRs

Consider the RR

$a_n = c_1 a_{n-1} + c_2 a_{n-2} \cdots + c_k a_{n-k} + f(n) \rightarrow (1)$

Let $k \geq 1$ be a positive integer. Then the order of RR (1) is k and its Corresponding degree is the highest power of terms $a_n, a_{n-1}, a_{n-2}, \cdots a_{n-k}$ in (1).

Example:-

Recurrence Relation	Order	Degree.
$a_n = a_{n-1} - a_{n-3}$	3	1
$a_n^2 = a_{n-1} - 100$	1	2
$\sqrt{a_n} = 2a_{n-1} - 3a_{n-2}$	2	2 (After squaring)
$\log(a_n) = na_{n-1} - n^2$	1	does not exists
$a_n = 2na^1_{n-1} a^1_{n-2}$	2	2 (1+1)

Linear and Non-linear recurrence Relation (10)

A Recurrence relation

$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + f(n)$

is called linear RR if its degree is 1. otherwise it is called non-linear RR.

Example:-

R.Rs	Degree	Linear/Non-linear
$a_n = a_{n-1} - n^2 a_{n-2} + a_{n-3}$	1	Linear RR
$a_n = a_{n-1} \cdot a_{n-2}$	2	Non-Linear RR
$a_n = 2a_{n-1} + n^2$	2	Non-Linear RR.

Homogenous and Non-Homogeneous RR's.

A recurrence relation

$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + f(n) \rightarrow$

is called (i) Homogeneous RR if $f(n) \equiv 0$

(ii) Non-Homogeneous RR if $f(n) \not\equiv 0$.

Example

R R	homogeneous/non-homogeneous.
$a_n = a_{n-1} + n$	Non-homogeneous ($\because f(n) = n$)
$a_n = n^2 a_{n-1} + 2$	Non-homogeneous ($\because f(n) = 2$)
$a_n = a_{n-1} + a_{n-2}$	Homogeneous ($\because f(n) = 0$.

(i) The given RR is

$a_n = 3a_{n-1} + 4a_{n-2} + 5a_{n-3}$

Its order is 3, degree is 1 and $f(n) = 0$.

$\therefore$ The given RR is

linear homogeneous RR

(ii) The given RR is

$a_n = 2n\, a_{n-1} + a_{n-2}$

The ~~given RR is~~

It's order is 2 $f(n) = 0$

degree is 1

$\therefore$ The given RR is linear homogeneous.

(iii) The given RR is

$$a_n = a_{n-1}^2 + a_{n-2}$$

It's order is 2, degree is 2, $f(n) = 0$

$\therefore$ The given RR is Non-linear homogeneous RR.

(5) Explain the general procedure of the characteristic equation method for solving linear homogeneous reccurrence relations with constant coefficients and apply this procedure to solve:

(i) $a_n = a_{n-1} + 2a_{n-2}$ (ii) $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ and (iii) $a_n = 6a_{n-1} - 9a_{n-2}$.

(6)(i) $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$

(ii) $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$.

(7) $f_n = f_{n-1} + f_{n-2}$, $f_0 = 0$, $f_1 = 1$

(8) (i) $a_n = 2a_{n-1} + 2a_{n-2}$ (ii) $a_n = a_{n-1} - a_{n-2}$.

Sol:- Characteristic Equation Method

Step 1: The given homogeneous linear recurrence relation is $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$ $\to$ (1)

Step 2: Consider the solution of (1) is

$$\boxed{a_n = A.r^n} \to (2)$$

where $A \neq 0$.

Step-3: Substituting (2) in (1)

$$A r^n = c_1.(A r^{n-1}) + c_2(A r^{n-2}) + \cdots + c_k (A r^{n-k})$$

$$\Rightarrow A r^n = A r^n [c_1 r^{-1} + c_2 r^{-2} + \cdots + c_k r^{-k}]$$

$$\Rightarrow 1 = c_1 r^{-1} + c_2 r^{-2} + \cdots + c_k r^{-k}$$

Dividing it by r^{-k}, we get

$$r^k = c_1 r^{k-1} + c_2 r^{k-2} + \cdots + c_k$$

$$\boxed{r^k - c_1 r^{k-1} - c_2 r^{k-2} \cdots - c_k = 0}$$ (12)

This is called ck equation of ①, and its roots $r_1, r_2, r_3 \ldots r_n$

Step 4: The following table discribes the nature of roots of ② and its corresponding solution a_n (or $a_n^{(h)}$) of ①

Nature of Roots of ③	Solution of ① (a_n or $a_n^{(h)}$)
* Roots are real and distinct $r_1, r_2, r_3 \ldots r_k$	$\longrightarrow a_n = A_1 r_1^n + A_2 r_2^n + \cdots A_k r_k^n$
* Roots are real and equal. $r_1 = r_2 = r$	$\longrightarrow a_n = (A_1 + nA_2) r^n$
$r_1 = r_2 = r_3 = r$	$\longrightarrow a_n = (A_1 + nA_1 + n^2 A_3) r^n$
* Roots are irrational $\alpha \pm \sqrt{\beta}$	$\longrightarrow a_n = A_1(\alpha+\sqrt{\beta})^n + A_2(\alpha-\sqrt{\beta})^n$
$\alpha \pm \sqrt{\beta}, r \pm \sqrt{s}$	$\longrightarrow a_n = A_3$
$\alpha \pm \sqrt{\beta}, \alpha \pm \sqrt{P}$	$\longrightarrow a_n = (A_1 + rA_2)(A + \sqrt{\beta})^n + (A_3 + nA_4)(\alpha - \sqrt{\beta})^n$
* Root are Complex numbers. $\alpha \pm i\beta$	$\longrightarrow a_n = A_1(\alpha_1 + i\beta) + A_2(\alpha - i\beta)^n$

5(i) The given RR is.

$a_n = a_{n-1} + 2a_{n-2} \longrightarrow$ ① by the method of CH-equation. Substi the Solution of ① is

$$\boxed{a_n = Ar^n} \longrightarrow ②$$

where $A \neq 0$

Substituting ② in ①, we get

$$Ar^n = Ar^{n-1} + 2(Ar^{n-2})$$

$$\Rightarrow Ar^n = Ar^n(r^{-1} + 2r^{-2})$$

$$\Rightarrow 1 = r^{-1} + 2r^{-2}$$

Divides it by r^{-2}, we get

$$r^2 = r + 2$$

$$\Rightarrow r^2 - r - 2 = 0$$

$$\Rightarrow (r+1)(r-2) = 0$$

$$\Rightarrow r = -1, 2 \text{ (Real and Distinct)}$$

the Solution of ① is

$$a_n = a_n^{(h)} = A_1 r_1^n + A_2 r_2^n$$

i.e $\boxed{a_n = A_1(-1)^n + A_2(2)^n}$

⑦ The given RR is

$$f_n = f_{n-1} + f_{n-2}, \quad f_0 = 0, \; f_1 = 1 \rightarrow ①$$

by the method of CK equation, the solution of ① is.

$$f_n = Ar^n \rightarrow ②$$

where $A \neq 0$

Substituting ② in ①, we get.

$$Ar^n = Ar^{n-1} + Ar^{n-1}$$

$$\Rightarrow 1 = r^{-1} + r^{-2}$$

Dividing it by r^{-2}, we get

$$r^2 = r + 1$$

$$\Rightarrow r^2 - r - 1 = 0$$

$$\Rightarrow r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2 \times a}$$

$$\Rightarrow r = \frac{(-1) \pm \sqrt{(-1)^2 - 4 \times 1 \times (-1)}}{2 \times 1}$$

$$\Rightarrow r = \frac{1 \pm \sqrt{5}}{2} \text{ (irrational root)}$$

$\therefore$ The Solution of ① is

$$f_n = A_1 r_1^n + A_2 r_2^n$$

i.e. $f_n = A_1\left(\frac{1+\sqrt{5}}{2}\right)^n + A_2\left(\frac{1+\sqrt{5}}{2}\right)^n \rightarrow ③$

Given that $f_0 = 0$

Put $n=0$ in ③, we get

$$f_0 = A_1\left(\frac{1+\sqrt{5}}{2}\right)^0 + A_2\left(\frac{1-\sqrt{5}}{2}\right)^0$$

$$\Rightarrow 0 = A_1 \times 1 + A_2 \times 1$$

$$\Rightarrow A_1 + A_2 = 0 \rightarrow ④$$

Given that $f_1 = 1$

Put $n=1$ in ③, we get

$$f_1 = A_1\left(\frac{1+\sqrt{5}}{2}\right)^1 + A_2\left(\frac{1-\sqrt{5}}{2}\right)^1$$

$$1 = A_1\left(\frac{1+\sqrt{5}}{2}\right) + A_2\left(\frac{1-\sqrt{5}}{2}\right), \text{ by } ④$$

$$1 = A_2\left[\frac{-\cancel{1}-\sqrt{5}+\cancel{1}-\sqrt{5}}{2}\right]$$

$$1 = A_2(-\sqrt{5})$$

$$\Rightarrow A_2 = \frac{-1}{\sqrt{5}} \quad \text{Similarly } A_1 = +\frac{1}{\sqrt{5}}$$

Substituting A_1 and A_2 in eq ③, we get the Complete Solution of ① is

$$\boxed{f_n = \frac{1}{\sqrt{5}}\left[1+\frac{\sqrt{5}}{2}\right]^n - \frac{1}{\sqrt{5}}\left[1-\frac{\sqrt{5}}{2}\right]^n}$$

(8) i) The given recurrence relation is

$$a_n = 2a_{n-1} - 2a_{n-2} \rightarrow ①$$

Consider the relation of ① as

$$a_n = a^{(h)} = Ar^n \rightarrow ②$$

where $A \neq 0$

from ① & ②, we get

$$1 = 2r^{-1} - 2r^{-2}$$

Dividing it by r^{-2}, we get

$$r^2 = 2r - 2$$

$$r^2 - 2r + 2 = 0$$

$$\Rightarrow r = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$\Rightarrow r = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times 1 \times 2}}{2 \times 1}$$

$$r = \frac{2 \pm \sqrt{4-8}}{2}$$

$$r = \frac{2 + \sqrt{-4}}{2}$$

$r = 1 \pm i$ (complex)

$\therefore$ The Solution of ① is

$$a_n = a_n^{(h)} = A_1(1+i)^n + A_2(1-i)^n.$$

8 (ii)

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(9) Explain the general procedure of the characteristic equation method for solving linear non-homogeneous recurrence relations with constant coefficients, and apply this method to solve:

(i) $a_n = 5a_{n-1} - 6a_{n-2} + 7^n$, and (ii) $a_n = 5a_{n-1} - 6a_{n-2} + 1$

Sol: Characteristics Equation method for non-homogeneous RR

Step (1): The given linear non-homogeneous RR is

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + f(n) \rightarrow ①$$

where $f(n) \neq 0$

Step (2): Consider the solution of ① as

$$\boxed{a_n = a_n^{(h)} + a_n^{(p)}} \rightarrow ②$$

where $a_n^{(h)}$ is the Solution of homogeneous part of ①

$a_n^{(p)}$ is the Solution of non-homogeneous part of ①

Step (3): To find $a_n^{(h)}$

Consider $\boxed{a_n^{(h)} = Ar^n}$ is the Solution of

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

$\therefore$ The equation.

$$\boxed{r^n - c_1 r^{n-1} - c_2 r^{n-2} - \cdots - c_k = 0}$$

is the Ch equation of ① and its roots are $r_1, r_2, \ldots r_k$.

By using table

$$a_n^{(h)} = A_1 r_1^n + A_2 r_2^n + \cdots + A_k r_k^n$$

Step 4: To find $a_n^{(p)}$

Consider the

$$\boxed{a_n^{(p)} = A.f(n)}$$

as the solution of particular part of ①

To calculate the value of A by using ①.

Step 5: Substituting $a_n^{(h)}$ and $a_n^{(p)}$ in ②

we get the solution a_n of ①

i) The given RR is

$$a_n = 5a_{n-1} - 6a_{n-2} + 7^n \rightarrow ①$$

To find $a_n^{(h)}$

Let $a_n^{(h)} = A r^n$. Then the homogenous part of ① is $a_n = 5a_{n-1} - 6a_{n-2}$ and its ch-equation is

$$1 = 5r^{-1} - 6r^{-2}$$

$$r^2 = 5r - 6$$

$$r^2 - 5r + 6 = 0$$

$$(r-2)(r-3) = 0$$

$$r = 2, 3$$

$$\boxed{a_n^{(h)} = A_1 2^n + A_2 . 3^n}$$

To find $a_n^{(p)}$

Let $a_n^{(p)} = A f(n) = A.7^n$ then from ① we have

$$A 7^n = 5(A 7^{n-1}) - 6(A 7^{n-2}) + 7^n$$

$$A 7^n = 7^n [5A 7^{-1} - 6A 7^{-2} + 1]$$

$$\Rightarrow A = \frac{5A}{7} - \frac{6A}{7^2} + 1$$

$$\boxed{A = \frac{49}{20}}$$

$$\boxed{a_n^{(p)} = \frac{49}{20} \cdot 7^n}$$

Hence, the solution of ① is

$$a_n = a_n^{(h)} + a_n^{(p)}$$

$$\boxed{a_n = A_1 2^n + A_2 3^n + \frac{49}{20} \cdot 7^n}$$

(ii) The given RR is

$$a_n = 5a_{n-1} - 6a_{n-2} + 1$$

To find $a_n^{(h)}$

Let $a_n^{(h)} = Ar^n$. Then the homogeneous part of ① is ~~an~~ $a_n = 5a_{n-1} - 6r^{-2}$

$\Rightarrow r^2 = 5r - 6$

$\Rightarrow r^2 - 5r + 6 = 0$

$\Rightarrow (r-2)(r-3) = 0$

$\Rightarrow r = 2, 3.$

$$\therefore \boxed{a_n^{(h)} = A_1 2^n + A_2 3^n}$$

To find $a_n^{(P)}$

Let $a_n^{(P)} = A f(n) = A.1 = A$ then from ① we have

$A = 5(A) - 6(A) + 1 \Rightarrow A = \frac{1}{2}$

$$a_n^{(P)} = \frac{1}{2}$$

Hence, the solation of ① is

$$\boxed{a_n = a_n^{(h)} + a_n^{(P)} \Rightarrow A_1 2^n + A_2 3^n + \frac{1}{2}}$$

⑩ Find the general solution of the non-homogeneous recurrence relations (i) $a_n = 3a_{n-1} + 2n$ and (ii) $a_n = 2a_{n-1} + 3^n$. and determine the particular Solution Satisfying $a_1 = 3$.

Sol: (i) The given RR is

$a_n = 3a_{n-1} + 2n \rightarrow$ ①

To find $a_n^{(h)}$

Let $a_n^{(h)} = Ar^n$. then the homogeneous part of ①

$a_n = 3a_{n-1}$ and its

CH equation is

(ii) The given RR is

$a_n = 2a_{n-1} + 3^n \longrightarrow$ ①'

To find $a_n^{(h)}$

Let $a_n^{(h)} = Ar^n$ then the homogeneous part of ①

$a_n = 2a_{n-1}$ & its CH

equation is

$1 = 3r^{-1} \Rightarrow A = 3 \;\; r = 3$

$\therefore$ The Solution is

$$\boxed{a_n^{(h)} = A_1 3^n}$$

To find $a_n^{(P)}$

Let $a_n = A \cdot f(n) = A \cdot (2n)$ be the Solution of ①. Then ① becomes

$$A(2n) = 3(A(2(n-1)) + 2n$$

$$\Rightarrow +2An = 6An - 6A + 2n$$

Equating coefficient of n on both sides.

$$2A = 6A + 2$$

$$\Rightarrow \boxed{A = -\frac{1}{2}}$$

$\therefore$ The Solution is,

$$\boxed{a_n^{(P)} = -\frac{1}{2}(2n)}$$

$\therefore$ The Complete Solution of ① is

$$a_n = a_n^{(h)} + a_n^{(P)}$$

$$\Rightarrow a_n = A_1 3^n + \left(-\frac{1}{2}(2n)\right)$$

$$a_n = A_1 3^n - \frac{1}{\cancel{2}}(\cancel{2}n)$$

$$\boxed{a_n = A_1 3^n - n} \rightarrow ③$$

Given that $a_1 = 3$

Taking $n = 1$ in ③. we get

$$a_1 = A_1 3^1 - 1$$

$$\Rightarrow 3 = 3A_1 - 1$$

$$\Rightarrow \boxed{A_1 = \frac{4}{3}}$$

From ③ we have

$$\boxed{a_n = \frac{4}{3}(3^n) - n}$$

$1 = 2r^{-1} \Rightarrow A = 2$

$\therefore$ The Solution is

$$\boxed{a_n^{(h)} = A_1 2^n}$$

(18)

To find $a_n^{(P)}$

Let $a_n = B \cdot f(n) = B \cdot 3^n$ be the solution of ①'. Then ①' can written as

$$B 3^n = 2(B 3^{n-1}) + 3^n$$

$$\Rightarrow B \cancel{3^n} = \cancel{3^n}[2B \cdot B^{-1} + 1]$$

$$\Rightarrow B = \frac{2B}{B} + 1$$

$$\Rightarrow \boxed{B = 3}$$

$\therefore$ The Solution is

$$\boxed{a_n^{(P)} = 3(3^n)}$$

The Complete Solution of ① is $a_n = a_n^{(h)} + a_n^{(P)}$

$$a_n = B_1 2^n + 3(3^n) \rightarrow ③'$$

Given that $a_1 = 3$

Taking $n = 1$ in ③'.

we get

$$a_1 = B_1 2^1 + 3(3^1)$$

$$\Rightarrow 3 = 2B_1 + 9$$

$$\Rightarrow B_1 = -3$$

$\therefore$ From ③' we have

$$\boxed{a_n = -3(2^n) + 3(3^n)}$$

(11) Express the Solution of the linear non-homogeneous recurrence relation $a_n = a_{n-1} + a_{n-2} + 1$, $n \geq 2$. with $a_0 = 0$, $a_1 = 1$, explicitly in terms of Fibonacci numbers

Sol: The given RR is

$$a_n = a_{n-1} + a_{n-2} + 1 \rightarrow ①$$

To find $a_n^{(h)}$

Let $a_n = A.r^n$ be the solution of homogeneous

$$\boxed{a_n = a_{n-1} + a_{n-2}}$$

Part of ①, then its CH equation is

$$1 = r^{-1} + r^{-2}$$

$$\Rightarrow r^2 - r - 1 = 0$$

$$\Rightarrow r = \frac{1 \pm \sqrt{5}}{2}$$

∴ The Solution is

$$a_n = A_1\left(\frac{1+\sqrt{5}}{2}\right)^n + A_2\left(\frac{1-\sqrt{5}}{2}\right)^n$$

To find $a_n^{(P)}$

Let $a_n = A.f(n) = A(1) = A$ be the solution of ①.

Then ① becomes

$$A = A + A + 1$$

$$\Rightarrow A = -1$$

∴ The Solution is

$$\boxed{a_n^{(P)} = -1}$$

Hence the Complete Solution of ① is

$$\boxed{a_n = A_1\left(\frac{1+\sqrt{5}}{2}\right)^n + A_2\left(\frac{1-\sqrt{5}}{2}\right)^n - 1} \Rightarrow a_n^{(h)} + a_n^{(p)}$$

Express the terms in ① as Fibonacci Sequence terms.

For this we consider.

$$\boxed{f_n = a_n + c} \rightarrow ②$$

From ① & ②, we get

$f_n - \cancel{c} = f_{n-1} - \cancel{c} + f_{n-2} - c + 1$ (20)

$\Rightarrow f_n = f_{n-1} + f_{n-2} + (c - c + 1)$

Taking $-c+1=0$

$\Rightarrow \boxed{c=1}$

From (2) we have

$\boxed{f_n = a_n + 1}$ (or) $\boxed{a_n = f_n - 1}$

where $f_0 = a_0 + 1 = 0 + 1 = 1$

$f_1 = a_1 + 1 = 1 + 1 = 2$

(12) Define the generating function of a sequence and illustrate with suitable examples. Determine the generating functions of the sequences:
(i) $\{3\}$, (ii) $\{n+1\}$, (iii) $\{2^n\}$, and (iv) $\{a^n\}$.

Sol:- **Generating Function:**

Let $\{a_n\}$ be the sequence of n^{th} term be a_n. Then its generating function is denoted by $G(x)$. And defined by

$$G(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n + \dots$$

(or)

$$G(x) = \sum_{x=0}^{\infty} a_n x^n$$

Example:-

Sequence	Generating Function
$\{1\}$	$G(x) = 1$
$\{1,1\}$	$G(x) = 1 + x$
$\{1,1,1\}$	$G(x) = 1 + x + x^2$
$\vdots$	$\vdots$
$\{1^n\}$	$G(x) = 1 + x + x^2 \dots \infty = \frac{1}{1-x}$
$\{a^n\}$	$G(x) = \frac{1}{1-ax}$
$\{-a^n\}$	$G(x) = \frac{1}{1+ax}$

(i) The given Sequence is $\{a_n\} = \{3\}$

Its generating function is

$\boxed{G(x) = 3}$

(ii) The given Sequence is $\{a_n\} = \{n+1\}$

GF is $G(x) = (1+0)+(1+1)x+(1+2)x^2+(1+3)x^3+\cdots$

$\Rightarrow 1+2x+3x^2+4x^3+\cdots$

$= (1-x)^{-2}$

(iii) The given Sequence is $\{a_n\} = \{2^n\}$

GF is $G(x) = 2^0+2^1x+2^2x^2+2^3x^3+\cdots$

$= 1+2x+(2x)^2+(2x)^3+\cdots$

$= (1-2x)^{-1}$

(iv) The given Sequence is $\{a_n\} = \{a^n\}$

GF is $G(x) = a^0+a^1x+a^2x^2+a^3x^3+\cdots$

$= 1+ax+(ax)^2+(ax)^3+\cdots$

$= (1-ax)^{-1}$

(13). Determine the closed-form of the sequence $\{a_n\}$ whose generating function is:

(i) $G(x) = \dfrac{x}{(1-2x)(1+3x)}$ (ii) $\dfrac{x+2}{(1-x)(1+x)}$ (iii) $\dfrac{1}{(1-ax)(1-bx)}$

Sol:- (i) The given GF is

$$G(x) = \frac{x}{(1-2x)(1+3x)}$$

By the partial-fractions, $G(x)$ can be written as

$$\frac{x}{(1-2x)(1+3x)} = \frac{A}{1-2x} + \frac{B}{1+3x}$$

where $A = \left(\dfrac{x}{1+3x}\right)_{x=1/2} = \dfrac{1/2}{1+3\left(\frac{1}{2}\right)} = \dfrac{1}{5}$

$B = \left(\dfrac{x}{1-2x}\right)_{x=-1/3} = \dfrac{-1/3}{1-2\left(-\frac{1}{3}\right)} = -\dfrac{1}{5}$

$\therefore$ The G.F can be written as (22)

$$G(x) = \frac{1/5}{1-2x} + \frac{-1/5}{1+3x}$$

$$= \frac{1}{5}(1-2x)^{-1} - \frac{1}{5}(1+3x)^{-1}$$

$$= \frac{1}{5}\sum_{n=0}^{\infty} 2^n x^n - \frac{1}{5}\sum_{n=0}^{\infty}(-3)^n x^n$$

Hence, the sequence of given G.F

$G(x) = \frac{x}{(1-2x)(1+3x)}$ is $\left\{\frac{1}{5}(2^n) - \frac{1}{5}(-3)^n\right\}$.

(ii) The given function is

$$G(x) = \frac{x+2}{(1-x)(1+x)}$$

By the partial-fractions, $G(x)$ can be written as

$$\frac{x+2}{(1-x)(1+x)} = \frac{A}{1-x} + \frac{B}{1+x}$$

where $A = \left(\frac{x+2}{1+x}\right)_{x=1} = \frac{1+2}{1+1} = \frac{3}{2}$

$B = \left(\frac{x+2}{1-x}\right)_{x=-1} = \frac{-1+2}{1-(-1)} = \frac{1}{2}$.

$\therefore$ The G.F can be written as

$$G(x) = \frac{3/2}{1-x} + \frac{1/2}{1+x}$$

$$= \frac{3}{2}(1-x)^{-1} + \frac{1}{2}(1+x)^{-1}$$

$$= \frac{3}{2}\sum_{n=0}^{\infty} 1^n x^n + \frac{1}{2}\sum_{n=0}^{\infty}(-1)^n x^n.$$

Hence, the sequence of given G.F $G(x) = \frac{x+2}{(1-x)(1+x)}$

is $\left\{\frac{3}{2}(1^n) + \frac{1}{2}(-1)^n\right\}$.

(iii) The given GF is,

$$G(x) = \frac{1}{(1-ax)(1-bx)}$$

By the partial fractions, it can be written as

$$\frac{1}{(1-ax)(1-bx)} = \frac{A}{1-ax} + \frac{B}{1-bx}$$

Where $A = \left(\frac{1}{1-bx}\right)_{x=1/a} = \frac{1}{1-b\left(\frac{1}{a}\right)} = \frac{a}{a-b}$

$$B = \left(\frac{1}{1-ax}\right)_{x=1/b} = \frac{1}{1-a\left(\frac{1}{b}\right)} = \frac{b}{b-a}$$

Now, given $G(x)$ can be written as

$$G(x) = \frac{\frac{a}{a-b}}{1-ax} + \frac{\frac{b}{b-a}}{1-bx}$$

$$= \frac{a}{a-b}(1-ax)^{-1} + \frac{b}{b-a}(1-bx)^{-1}$$

$$= \frac{a}{a-b}\sum_{n=0}^{\infty} a^n x^n + \frac{b}{b-a}\sum_{n=0}^{\infty} b^n x^n.$$

Hence, the sequence of given GF $\frac{1}{(1-ax)(1-bx)}$ is $\left\{\frac{a}{a-b}(a^n) + \frac{b}{b-a}(b^n)\right\}$.

Generating Function method for solving RR.

Step (1): The given RR is

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + f(n) \rightarrow (1)$$

Step (2). Replacing n by $n+k$ in (1) we get

$$a_{n+k} = c_1 a_{n+k-1} + c_2 a_{n+k-2} + \cdots + c_k a_n + f(n) \rightarrow (2)$$

Step (3). Taking

$$G(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n + \cdots$$

$$\therefore \frac{G(x)-a_0}{x} = a_1 + a_2 x^2 + \cdots + a_n x^n + \cdots$$

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$$\Rightarrow \frac{G(x)-a_0}{x} = \sum_{n=0}^{\infty} a_{n+1} x^n.$$

Similarly, we can determine

$$\frac{G(x)-a_0-a_1 x}{x^2} = \sum_{n=0}^{\infty} a_{n+2} x^n.$$

$$\frac{G(x)-a_0-a_1x-a_2x^2}{x^3} = \sum_{n=0}^{\infty} a_{n+3} x^n;$$

$\vdots$

$$\frac{G(x)-a_0-a_1x-a_2x^2-\cdots-a_{k-1}x^{k-1}}{x^k} = \sum_{n=0}^{\infty} a_{n+k} x^n$$

$\vdots$

step(3): Substituting $a_n, a_{n+1}, a_{n+2}, \ldots, a_{n+k}$ in terms of $G(x)$ in ②, we get the required GF $G(x)$ in terms of x.

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(14) prove that the GF of the Fibonacci sequence is $G(x) = \frac{x}{1-x-x^2}$, and use partial fractions to recover closed form of the corresponding Fibonacci sequence.

Sol: The recurrence relation for the Fibonacci sequence $\{f_n\}$ is

$$f_n = f_{n-1} + f_{n-2} \longrightarrow ①$$

where $f_0 = 0$, $f_1 = 1$.

Replacing n by $n+2$ in ① we get

$$f_{n+2} = f_{n+1} + f_n \longrightarrow ②$$

applying summation both sides of ② from $n=0$ to $n=\infty$ with the multiplication of x^n, we get

$$\sum_{n=0}^{\infty} f_{n+2} x^n = \sum_{n=0}^{\infty} f_{n+1} x^n + \sum_{n=0}^{\infty} f_n x^n$$

$$\Rightarrow \frac{\varphi(x) - f_0 - f_1 \cdot x}{x^2} = \frac{\varphi(x) - f_0}{x} + \varphi(x)$$

$$\Rightarrow \varphi(x) - f_0 - f_1 \cdot x = x\varphi(x) - x \cdot f_0 + x^2 \varphi(x)$$

$$\Rightarrow \varphi(x)[1 - x - x^2] = f_0 + f_1 \cdot x - x \cdot f_0$$

Given that $f_0 = 0$, $f_1 = 1$

$$\Rightarrow \varphi(x)[1 - x - x^2] = x.$$

$$\Rightarrow \boxed{\varphi(x) = \frac{x}{1 - x - x^2}}.$$

We know that $\alpha = \frac{1+\sqrt{5}}{2}$ and $\beta = \frac{1-\sqrt{5}}{2}$ be the roots of $x^2 - x - 1 = 0$. Then it can be written as

$$\boxed{1 - x - x^2 \equiv (1 - \alpha x)(1 - \beta x)}.$$

$$\therefore \varphi(x) = \frac{x}{(1-\alpha x)(1-\beta x)} \equiv \frac{\frac{1}{\alpha - \beta}}{1 - \alpha x} + \frac{\frac{1}{\beta - \alpha}}{1 - \beta x}.$$

$$\Rightarrow \varphi(x) = \frac{1}{\alpha - \beta}(1 - \alpha x)^{-1} + \frac{1}{\beta - \alpha}(1 - \beta x)^{-1}.$$

$$\Rightarrow \varphi(x) = \frac{1}{\alpha - \beta} \sum_{n=0}^{\infty} \alpha^n x^n + \frac{1}{\beta - \alpha} \sum_{n=0}^{\infty} \beta^n x^n.$$

Hence, the solution of (1) is

$$\{f_n\} = \left\{ \frac{1}{\alpha - \beta}(\alpha^n) + \frac{1}{\beta - \alpha}(\beta^n) \right\}$$

i.e., $$\{f_n\} = \left\{ \frac{\alpha^n - \beta^n}{\alpha - \beta} \right\}$$

(15) solve (i) $a_n = 3a_{n-1}, a_0 = 2$

(ii) $a_n = 8a_{n-1} + 10^{n-1}, a_0 = 9$ by generating function method.

Sol:

(i) The given RR is

$a_n = 3a_{n-1} \rightarrow (1)$

with $a_0 = 2$

Replacing n by n+1 in (1) we get

$a_{n+1} = 3a_n \rightarrow (2)$

Applying summation on both sides of (2) with multiplication x^n on both sides of (2),

$\sum_{n=0}^{\infty} a_{n+1} x^n = 3 \sum_{n=0}^{\infty} a_n x^n \rightarrow (3)$

By the method of GF,

$G(x) = \sum_{n=0}^{\infty} a_n x^n$

$\frac{G(x) - a_0}{x} = \sum_{n=0}^{\infty} a_{n+1} x^n.$

From (3) we have

$\frac{G(x) - a_0}{x} = 3G(x)$

$\Rightarrow G(x) - 2 = 3x\, G(x)$

$\Rightarrow G(x)[1 - 3x] = 2$

$\Rightarrow G(x) = \frac{2}{1-3x}$

$= 2(1-3x)^{-1}$

$= 2\sum_{n=0}^{\infty} 3^n x^n.$

$\therefore$ the solution of (1) is

$\{a_n\} = \{2 \cdot 3^n\}$

(ii) The given RR is

$a_n = 8a_{n-1} + 10^{n-1} \rightarrow (1)$

Replacing n by n+1 in (1),

$a_{n+1} = 8a_n + 10^n \rightarrow (2)$

Applying summation

$\sum_{n=0}^{\infty} a_{n+1} x^n = 8\sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} 10^n x^n \rightarrow (3)$

By the method of GF,

$G(x) = \sum_{n=0}^{\infty} a_n x^n,$

(3) becomes

$\frac{G(x) - a_0}{x} = 8G(x) + \frac{1}{1-10x}$

$\Rightarrow G(x) - 9 = x8 \cdot G(x) + \frac{x}{1-10x}$

$\Rightarrow G(x)[1 - 8x] = 9 + \frac{x}{1-10x}$

$\Rightarrow G(x) = \frac{9}{1-8x} + \frac{x}{(1-8x)(1-10x)}$

But

$\frac{x}{(1-8x)(1-10x)} = \frac{-1/2}{1-8x} + \frac{+1/2}{1-10x}.$

$\therefore G(x) = \frac{9}{1-8x} + \frac{-1/2}{1-8x} + \frac{1/2}{1-10x}$

$= 9(1-8x)^{-1} - \frac{1}{2}(1-8x)^{-1} + \frac{1}{2}(1-10x)^{-1}$

$= 9\sum_{n=0}^{\infty} 8^n x^n - \frac{1}{2}\sum_{n=0}^{\infty} 8^n x^n + \frac{1}{2}\sum_{n=0}^{\infty} 10^n x^n$

$\therefore$ the solution of (1) is

$\{a_n\} = \{9 \cdot 8^n - \frac{1}{2} \cdot 8^n + \frac{1}{2} \cdot 10^n\}$